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REVIEW ARTICLE

Development of food and drug homologous products: insights from raw material, modern processing technologies and quality assurance

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Abstract
In recent years, there has been an observed increase in health issues stemming from poor diets and inactivity. The utilization of homologous products, which offer dual nutrition and functionality, demonstrates considerable promise in the domains of prevention and treatment. Nevertheless, its potential remains largely underutilized. A significant constraint pertains to the predominance of individual technologies in the current research focus, with a paucity of systematic integration across the entire supply chain. To address this gap, the present paper undertakes a systematic review of the raw materials, followed by an elucidation of the application technologies in this field, including advanced processing technologies, analytical identification technologies and quality and safety control technologies. This paper finally explores the development challenges and potential opportunities. It also provides a cohesive knowledge framework to underpin their standardized production and industrialization, bridging the gap between academic research and practical application.

Keywords Functional food · Food and drug homologous products · Raw materials · Key technologies · Health intervention

Introduction

Over the past few decades, ultra-processed junk food and fast food have become increasingly common worldwide, contributing significantly to the rise in diet-related non-communicable diseases such as obesity and diabetes [95]. Despite being aware of the harms of an unhealthy diet, many people still opt for convenient foods over healthier options due to busy schedules and lifestyles [20]. This creates a gap between the idea of preventing diseases through diet and actual eating habits. However, there is a notable increase in consumer demand for functional foods, which are enriched with various bioactive ingredients at effective doses [85]. These foods offer benefits beyond basic nutrition, aiding in disease prevention and management [12]. Studies indicate that 68% of global consumers now buy functional foods to address specific health concerns [26], suggesting a growing awareness of using food for diseases prevent [45].

Compared to ordinary functional foods, food and drug homologous products offer a key advantage in their naturalness and safety, grounded in traditional health concepts. These products do not need additional artificial bioactive ingredients, instead, they leverage the natural properties of food ingredients to provide regulatory effects [10]. Using natural raw materials that function both as food and medicine, they address concerns about food safety while aligning with the current trend of maintaining health through daily diets [129]. These products can be used directly in cooking (such as porridge with Chinese yam) or processed into convenient ready-to-eat items (e.g., wolfberry original pulp, Poria cakes). This approach effectively balances health needs with a convenient lifestyle, meeting the modern consumer's desire to prevent diseases through diet without adding to daily burdens [11].

This paper systematically explores the raw material characteristics, application technologies, and industrial development challenges of food and drug homologous products. First, it sorts out the core raw material categories

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and analyzes their corresponding functional attributes and mechanism of action. Second, it focuses on three key technologies (advanced processing, analytical identification and quality and safety control), then deeply explains the core principles, applicable scenarios, advantages and disadvantages of each type of technology. Finally, it analyzes the current practical dilemmas and potential development opportunities in the field, puts forward a forward-looking development vision. This provides theoretical support and practical guidance for the standardized production, functional optimization and industrial application of food and drug homology products.

Raw materials for food and drug homologous products

Plant raw materials

There is an abundant and diverse range of botanicals for food and drug, which can be divided into four categories. Seasonings, including cinnamon, licorice, cloves, fennel, star anise, *Angelica dahurica*, turmeric, etc. Daily foods, including yam, taro, hawthorn, etc. Tea drinks, including bitter orange flower, honeysuckle, chrysanthemum, goji berry, poria cocos, lotus leaf and lotus seed, etc. Other dietary and health supplements, including black sesame, dandelion, eucommia ulmoides leaves, astragalus, Lingzhi, and Codonopsis pilosula, etc. (Li et al., 2024).

In the seasoning category, in addition to serving as daily seasonings, these substances are also used in the research and development of functional foods. For instance, blood sugar controlling biscuits are developed with cinnamon as the core raw material, which utilize cinnamaldehyde contained in cinnamon to assist in regulating postprandial blood sugar and thus cater to consumer groups who need to control their blood sugar [46]. Glycyrrhetinic acid and other active ingredients extracted from licorice are added to throat soothing soft candies to exert the effect of relieving throat discomfort [139]. Dahiru et al. [21] demonstrated that clove-derived polyphenols exert dose-dependent protective effects against cell oxidative damage: when the polyphenol concentration reaches 10 mg/L, they significantly mitigating oxidative stress-induced cell injury. In terms of antibacterial activity, Devi et al. [23] found that eugenol can alter the integrity of the cell membrane of Proteus mirabilis and affect the physiological activity of the cells, thereby achieving antibacterial effects. In functional foods, clove is mostly developed around three core functions: gum protection and teeth strengthening, stomach invigoration and digestion promotion and bacteriostasis and preservation. Fermented tea made from clove leaves contains eugenol, which can inhibit the activity of *Helicobacter pylori*. Additionally, it contains components such as theaflavins and thearubigins, which help promote gastrointestinal digestion and stimulate appetite [2].

Among the daily foods, in addition to being used as daily food ingredients, yam, taro and hawthorn are also applied in the research and development of functional foods. For example, spleen-invigorating and stomach-nourishing meal replacement porridge is made with yam as the main raw material, which uses the yam polysaccharides it contains to help regulate gastrointestinal function and is suitable for people with weak digestive function [22]. Taro is mixed with whole grain flour to develop high-fiber satiating biscuits, which leverage the dietary fiber in taro to extend the feeling of fullness and meet the needs of people who want to control their weight [81]. Active ingredients such as flavonoids and organic acids extracted from hawthorn are added to digestive and stagnation-relieving chewable tablets to play a role in promoting digestion and relieving postprandial abdominal distension [42]. The organic acids in hawthorn also exhibit significant therapeutic effects, such as lowering blood pressure, regulating blood lipids, preventing tumors, and exerting anti-inflammatory activity [115].

In addition to being used in tea beverages, honeysuckle and chrysanthemum are also applied in the research and development of functional health products. For example, chlorogenic acid in honeysuckle has a significant antibacterial effect, which can effectively inhibit Salmonella typhi, Staphylococcus aureus, Pneumococcus and Mycobacterium tuberculosis, has been widely extracted and used in clinical treatment [140]. Honeysuckle is compounded with mint extract to make heat-clearing and throat-moistening sprays, which leverage the bacteriostatic effect of honeysuckle and the coolness of mint to quickly relieve sore throat and dry discomfort [44]. Camphor in chrysanthemum volatile oil has antibacterial and anti-inflammatory effects. Pharmacological analysis has shown that chrysanthemum extract can enhance capillary resistance, inhibit capillary permeability, and have anti-inflammatory effects [65]. In some functional foods such as ready-to-eat fruits and vegetables and soybean products, a low concentration of chrysanthemum extract is added. Its antibacterial components can delay the reproduction of harmful microorganisms in the food, extend the shelf life, while preserving the natural properties of the food and reducing the use of chemical preservatives [99].

Other dietary and health supplements are also frequently used in the production of functional foods. For instance, black sesame oil and sesamin are extracted from black sesame and then blended with whole grain flour to create blackening and hair-nourishing meal replacement powder, which, by virtue of its unsaturated fatty acids and nutrients, helps alleviate issues such as dry and brittle hair [78]. Li

et al. [64] summarized that dandelion products and derived constituents exhibit pharmacological effects against GI disorders. Meanwhile, taraxasterol and flavonoids extracted from dandelion are added to heat-clearing and diuretic solid beverages, and its heat-clearing property aids in relieving discomforts like dry mouth and difficult urination caused by internal dampness and heat [69]. Ganoderma lucidum is a promising source of prebiotics due to its abundance of several bioactive compounds that have nutritional and medicinal effects and are present in all parts of the fungus [98]. A variety of beverages made from Ganoderma lucidum, such as Ganoderma lucidum tea and functional drinks, have functions including enhancing immunity and relieving fatigue [114].

Animal Raw materials

Animal raw materials similar to food and drugs can be classified as insects, such as honey and propolis made by bees. Birds, such as swiftlets, which use saliva to build nests. Mammals, such as pilose antler made from the young horns of the sika deer, donkey-hide gelatin made of donkey skin, etc.

In the category of insects, honey and propolis not only supplement nutrition [68], relieve pain and detoxify, but also have anti-inflammatory, antibacterial, and maintenance of intestinal microbiota homeostasis effects. The confectionery derived from honey and propolis has been demonstrated to possess a variety of therapeutic benefits, including the treatment of oral diseases, throat ailments, halitosis, and digestive tract disorders. In addition to these substantial benefits, the product contains the nutritional components inherent in natural honey and has been shown to possess antiviral properties (Rodica, Erkan et al. [90]). Rani, Budumuru, and Bandaru [86] have shown that honey has a significant inhibitory effect on Staphylococcus aureus, with flavonoids and phenolic acids being the main antibacterial active substances. Silva-Beltrán et al. [101] demonstrated through in vitro studies that Mexican brown propolis extract Brazilian green and red propolis extract, and their phenolic compounds (quercetin, caffeic acid, and rutin) have antiviral effects in preventing HCoV-229E infection of normal human lung fibroblasts.

Among the birds' category, bird's nest not only has the function of nourishing and beautifying the skin (Yuan, Lin, Wang, & Dai [134]),, but also has functions such as immune regulation, anti-inflammatory, promoting the intellectual development of the offspring, and neuroprotection. The sialic acid contained in bird's nest can neutralize the influenza virus, thereby preventing the invasion of the virus and cell surface sialic acid receptors. Hou et al. [40] showed that bird's nest is rich in lactoferrin and ovotransferrin, which effectively promote neuronal protection by scavenging free radicals in SH-SY5Y cells. Instant bird's nest or bird's nest congee is widely loved by people for its efficient nourishment and convenient consumption.

In the category of Mammals, deer antler contains various amino acids, phospholipids, hormones, etc., which have anti-fatigue, anti-osteoporosis, and immune-enhancing effects [57]. Zhang et al. [136] showed that the nucleosides in deer antler can significantly enhance the anti-fatigue ability of mice. Widyowati et al. [121] found that both alcohol and water extracts of deer antler had protective effects on dexamethasone induced osteoporosis in mice. Deer antler health wine is deeply loved by people for its advantages such as relieving fatigue and warming the body. Donkey-hide gelatin is rich in collagen, which has the functions of beauty, anti-inflammatory, antibacterial and anti-aging. The study found that donkey-hide gelatin peptide iron chelate can significantly improve the colonic mucosal ulcer, inflammatory cell infiltration and reduce oxidative stress in anemic mice. Compared with traditional iron supplements, high doses of donkey-hide gelatin peptide iron chelate do not lead to an increase in pathogenic bacteria and a decrease in beneficial bacteria. It has higher safety and can more stably maintain the balance of the intestinal environment [17]. [127] found that donkey-hide gelatin has an inhibitory effect on skin aging caused by hydrogen peroxide and ultraviolet light.

Food and drug homologous products technologies

Extraction, purification, and fermentation technologies

As shown in Table 1, advanced processing technologies for food and drug homologous products include extraction, purification and fermentation, which are crucial for improving product quality and efficacy. The application of these advanced processing technologies not only enhances the market competitiveness, but also injects new vitality into the development of the large healthcare industry.

Extraction technology

Common extraction technologies for bioactive ingredients in food and drug homologous products mainly include conventional extraction technologies and advanced extraction technologies. Conventional extraction technologies, represented by solvent extraction (e.g., ethanol extraction, water extraction), are widely used in industrial production due to their mature process, low equipment investment, and strong

Table 1 Advanced processing technologies such as extraction, purification, and fermentation of food and drug homologous

Technologies	Sources	Principles	Advantages	Disadvantages	References
Solvent extraction method	Almond skins, Centella asiatica, Oyster.	Extracting effective ingredients from raw materials using appropriate solvents.	Easy operation, low equipment investment.	long extraction time, high energy consumption and potential residual organic solvents.	[104]
Supercritical fluid extraction	Ginger, Fritillaria thunbergia, Calendulacea.	Selective extraction of active ingredients from raw materials using supercritical fluids (such as supercritical CO_2) as solvents.	Process simple, high efficiency, low energy, mild conditions, pollution-free, residue-free.	High equipment investment.	[24]
Low-eutectic solvents extraction method	Chinese wolfberry, Honeysuckle, Ginseng, astragalus.	Its unique molecular structure interacts with target extracts, enhancing the latter's solubility in the solvent.	Low toxicity, cost-efficiency and a wide range of solubility.	High cost, limited applicability.	[88]
Enzyme-assisted extraction	Turmeric, Ganoderma lucidum.	By the specific catalytic action of enzymes, the cell wall and membrane are broken down, thereby releasing the active ingredients inside the cell.	Efficient, gentle, low pollution, low cost.	Affected by raw materials and environment.	[47]
Ultrasonic-assisted extraction	Hawthorn, taro, ginger.	Utilization of high frequency mechanical waves to accelerate the rupture of plant cells and tissue structures while enhancing solvent penetration.	High efficiency and short time consuming.	Excessive power may lead to degradation of the extracted substance and decrease in yield.	[138]
Microwave -assisted method	Black pepper, Loquat leaf polysaccharides, Dioscorea hispida.	Cells are ruptured and their contents are released under the pressure of a magnetic field.	Short time consumption, with the characteristics of high efficiency, energy saving, and time-saving.	Low yield, uneven heating.	[93]
Infrared-assisted extraction	Orange peel, *Saussurea lappa* L, Pomegranate.	Infrared rays, via their thermal effect, damage the cell wall structure of the material.	High extraction rates, short processing times, simple operation and reduced energy consumption.	Limited penetration and high equipment cost	[125]
Pulsed electric field technology	Ganoderma lucidum, Almond, Ginseng, Honeysuckle.	Accelerating the dissolution of target components by utilizing the electroporation effect.	Higher product yields and shorter extraction times.	Influenced by the specific conditions	[7]
Cold plasma treatment	Radix Astragali, Radix Glycyrrhizae.	It generates reactive species through electrical discharge, which can break down the structure of raw materials.	Higher energy efficiency, less chemical reagent usage, less waste.	High initial equipment cost and limited raw material scope.	[59]
Macroporous adsorption resin technology	Honey, Lycium barbarum L.	It achieves separation and purification by adsorbing organic compounds in solution selectively.	Low cost, high product purity, no moisture absorption, and reusable, easily regenerable.	Limited adsorption selectivity.	[34]
Chromatographic Separation and Purification Technology	Radix Ginseng, Radix Astragali, Ganoderma lucidum crude polysaccharides.	It achieves separation and purification by differential migration of substances.	Good selectivity, high separation efficiency, and wide applicability.	High equipment maintenance costs.	[80]
Membrane separation technology	Ginsenosides, Chrysanthemum, Ganoderma lucidum, Donkey-hide gelatin, Oyster.	Selective separation of molecules with different particle sizes in a mixture using special thin films.	Simple equipment, no phase change, energy-saving and efficient, no secondary pollution.	Limited lifespan and ability to withstand extreme conditions.	[87]
Fermentation technology	Wolfberry, Mulberry, Cornus.	Through the metabolic process of microorganisms, the effective ingredients in the product are better released and converted.	Short process cycle, controllable production, stable quality.	Influenced by various environmental factors.	[110]

operability, but they generally have shortcomings such as long extraction time, high energy consumption, and potential residual organic solvents [104]. Advanced extraction technologies have been developed to address these limitations, with supercritical fluid extraction, low-eutectic solvents, enzyme-assisted extraction, ultrasound-assisted extraction, microwave-assisted extraction, infrared-assisted extraction, pulsed electric field technology, cold plasma treatment, etc. being the most commonly studied and applied types in recent years [29].

The background of the solvent extraction method stems from the demand for substance separation and purification in industrial production, and it was initially developed to address the issue of efficiently separating target components in fields such as ore smelting and chemical raw material extraction. With the advancement of science, this method has gradually expanded to various industries [77]. Solvent extraction method is the use of appropriate solvents to extract active ingredients from raw materials [61]. This method is simple and easy to implement and is one of the traditional methods for extracting food and drug homologous substances. There are many solvent extraction methods, such as hot water extraction, alkaline aqueous solution method, ethanol extraction method, etc [49]. These methods reduce the threshold for production equipment and operating costs in the food and drug industries.

The development of supercritical fluid extraction (SFE) technology was driven by the necessity for efficient and environmentally friendly separation methods. Over the past few decades, the scope of applications for SFE has expanded considerably [119]. SFE is the selective extraction of active ingredients from raw materials using a fluid in a supercritical state (such as supercritical CO_2) as a solvent [30]. Wu et al. [122] established off-line combined method of SFE and supercritical fluid chromatography for the flavonoid components in ginger. Under the process conditions (co-solvent 8% ethanol, temperature 45 °C, pressure 30 MPa, reaction time 30 min), the extraction of the target components was successfully achieved, and three high-purity flavonoid compounds were obtained. Ruan et al. [91] were the first to use the supercritical fluid extraction technique to extract total alkaloids from the flowers of *Fritillaria thunbergii*, saving a lot significant time and resource savings. The extraction capability of supercritical fluid extraction technology is determined by the density of the fluid, which can be controlled by adjusting the temperature and pressure. It is easy to control and has the advantages of no pollution, no contamination, etc., which ensures stable product quality [24]. This innovative methodology presents several limitations, notably the substantial capital investment necessary for establishing and equipping SFE facilities, rendering it financially impractical for smaller laboratories and research institutes [93].

The core principle of extraction using low-eutectic solvents (LESs) is: leveraging their unique molecular structure (containing hydrogen bond donors and acceptors) to form hydrogen bonds, electrostatic interactions or hydrophobic interactions with target extracts, thereby enhancing the solubility of the target ingredients in the solvent. At the same time, LESs disrupt the cell structure of raw materials, ultimately achieving the efficient transfer and separation of target ingredients from the raw materials to the LESs system [142]. Compared with traditional extraction methods, low-eutectic solvents have emerged as an eco-friendly alternative, offering advantages such as low toxicity, cost-efficiency and a wide range of solubility. The extraction of bioactive compounds from apple pomace reported that the use of LESs has resulted in a higher yield of total phenolic compounds [88]. Other than that, Shafie and Gan. (2019) reported that the extraction of polysaccharide from *Averrhoa bilimbi* fruits by using choline chloride–citric acid monohydrate-based low-eutectic solvents produced a higher yield than conventional extraction solvent. In the future, refining low-eutectic solvents-based extraction processes and gaining a deeper understanding of their molecular dynamics will be crucial in enhancing their industrial use and accelerating the transition toward more sustainability.

Enzyme-assisted extraction (EAE) is an environmentally friendly, nonthermal technique introduced in the last decade. It uses cellulases, pectinases, hemicellulases and their combination to break down the food matrixes for better extraction of polyphenols, carotenoids, essential oils and proteins, etc. from natural sources compounds [13]. The enzyme extraction method involves the specific catalytic action of enzymes to break down the cell wall and membrane, thereby releasing the active components within the cell [75]. Enzymatic extraction can selectively disrupt the cell structure and improve the release efficiency of active ingredients [131], it is usually carried out at room temperature and pressure, to avoid the denaturation or loss of active ingredients that may be caused by high temperature and pressure. The use of fewer solvents reduces environmental pollution. In addition, enzymatic extraction can be combined with other extraction methods to form composite extraction techniques, further improving extraction efficiency and purity of active ingredients.

The development of ultrasonic-assisted extraction (UAE) relies on the research into the physical properties of ultrasonic waves; these properties can enhance mass transfer processes, thus gradually enabling the technology to be applied in the extraction of target active components in fields such as food and pharmaceuticals [58]. UAE uses high-frequency mechanical waves to generate strong high-speed jets or

shock waves, forming high-frequency vibrations, generating cavitation and thermal effects, thereby accelerating the deformation and fracture of plant cells, tissues and structures, accelerating solvent penetration into raw materials [94], improving extraction efficiency, and shortening extraction time [113]. Chen et al. [15] used water immersion extraction, ultrasonic-assisted extraction, enzyme-assisted extraction and ultrasonic-enzyme-assisted extraction to extract polysaccharides from hawthorn. Among them, the hawthorn polysaccharides extracted by ultrasonic-assisted extraction had the best superoxide radical scavenging ability. Taro polysaccharides have antioxidant, lipid-lowering and immunomodulatory effects. Compared with the water extraction and alcohol precipitation methods, the ultrasound-assisted method has a lower temperature, which can reduce the damage of high temperature to the active ingredients. At the same time, UAE has a shorter extraction time, which can effectively improve the extraction efficiency [38].

In recent years, microwave-assisted extraction (MAE) technology has been widely applied to extract target components such as polyphenols, flavonoids and alkaloids [19]. MAE uses short-wavelength electromagnetic waves to rupture cells and release their contents with the dielectric heating, which fuse and diffuse with the extraction solution during immersion [83]. Compared with direct water extraction or alcohol precipitation, MAE has the advantages of shorter processing times, higher efficiency and reduced energy consumption. The reason may be that MAE has a shorter extraction time and suitable temperature, which reduces the damage of high temperature to the extracted material. However, the extraction rate is lower and needs to be further adjusted [123]. Piperine is derived from black pepper and has effects such as anticancer, anticonvulsant, antidepressant, anti-inflammatory, antioxidant, immune regulation, and anti-obesity. Fu et al. [33] investigated the extraction of loquat leaf polysaccharides using hot water extraction, pressurized water extraction, high-speed shear homogenization extraction, MAE, UAE, ultrasound-assisted enzymatic extraction, and ultrasound-microwave-assisted extraction. The results showed that polysaccharides extracted from loquat leaves by MAE and ultrasound-assisted enzymatic exhibited strong antioxidant activity. This green method has some limitations. MAE can sometimes cause selective heating of certain substances, leading to uneven heating. It is also more expensive and its operations can be more challenging, especially when dealing with uneven samples [93].

The characteristics of infrared radiation have paved the way for the development of new extraction technologies. Infrared rays, via their thermal effect, cause rapid energy absorption and intense vibration of water molecules or target components inside the material and further damage the cell wall structure of the material, reduces the binding force between target components and the matrix, and ultimately accelerates the diffusion of target components into the extraction solvent. Based on these characteristics, infrared-assisted extraction (IAE) technology has emerged as a result [9]. This technology offers advantages such as high extraction rates, short processing times, simple operation and reduced energy consumption [125]. El Kantar et al. [28] applied IAE technology to extract polyphenols from orange peel. Compared to untreated peel, infrared treatment improved polyphenol extraction efficiency. Raafat et al. [84] employed IAE to extract lactones from *Saussurea lappa* L. They found that the lactone yield from the infrared-assisted extraction method was 1.6 times and 2.7 times higher than that from the ultrasonic-assisted extraction and solid-liquid extraction methods, respectively. IAE has advantages such as high heating efficiency, targeted action on active components to reduce damage to bioactive substances, no operational noise and wide adaptability to extraction media, but it also has disadvantages including limited penetration of infrared radiation and relatively high equipment costs for large-scale applications [72].

The core application of pulsed electric field (PEF) technology in food enterprises mainly focuses on the processing of liquid and semi-liquid foods. PEF works on the principle of electroporation: by applying electrical pulses to food, pores are created in the cell membranes of raw materials. These pores form channels for the movement of molecules, ions, and water, thereby accelerating the dissolution of target components [51]. This characteristic not only improves the efficiency of foods pressing and extraction but also facilitates the release and retention of intracellular substances such as flavor compounds, natural pigments, and active nutrients, thereby enhancing the flavor intensity, nutritional content, and appearance color of the final products. PEF employs mild energy treatment that facilitates a specific extraction of high-purity intracellular compounds compared to other treatments such as ultrasonication, microwave-assisted extraction and high-voltage electric discharge [96]. Sulforaphane is an isothiocyanate mostly occurring in cruciferous vegetables; it possesses anti-cancer properties and may reduce the risk of cardiovascular diseases (Vanduchova, Anzenbacher, & Anzenbacherova [109]),. And it was observed that the recovery of sulforaphane was increased with electric field intensity and exposure time [71]. PEF enables higher product yields and shorter extraction times compared to conventional extraction approaches. But the effectiveness of the method is strongly influenced by the specific conditions of the extraction process (e.g., pulse intensity, treatment time, raw material properties), requiring precise parameter optimization and limiting its adaptability to some scenarios [7].

Currently, the food, drug and cosmetic industries have an urgent demand for green extraction. They need to not only preserve the biological activity of extracts, but also reduce energy consumption and pollution. Against this backdrop, cold plasma (CP) technology has emerged. Operating at near-ambient temperatures, it generates reactive species through electrical discharge, which can break down the structure of raw materials (e.g., plant cell walls) without damaging heat-sensitive components. This makes it suitable for extracting high-value natural substances [112]. CP can cause cell disruption due to reactive species generation and enhance solvent penetration; thereby, it helps in improving extraction yield with negligible effects on characteristics [59]. In contrast to traditional methods, such as steam distillation that depends on heat, CP operates at room temperature or slightly above, which is crucial for preserving sensitive and volatile compounds like essential oils, polyphenols, antioxidants, and vitamins [6, 56]. Total phenols have multiple effects, including antioxidant, anti-inflammatory, and metabolic regulation, as well as cardiovascular protection, which are important active ingredients for maintaining human health. Kuo-Ching Jan and Mohsen Gavahian [60] found in their study that cold plasma treatment could increase the total phenol content of blue pea powder and blue pea aqueous water extract compared with the control treatment. CP has advantages including higher energy efficiency, less chemical reagent usage, less waste, and the ability to retain component activity. Its disadvantages are high initial equipment cost and limited raw material scope.

Purification technology

Purification technologies for extracts of food and drug homologous products are mainly divided into traditional purification technologies and modern advanced purification technologies. Traditional purification technologies include methods such as solvent precipitation and resin adsorption [82]. Solvent precipitation is widely used for preliminary purification of polysaccharides and proteins due to its simple operation and low cost. However, it has limitations such as low purification efficiency and potential residual organic solvents [74]. Resin adsorption, especially macroporous resin adsorption, is suitable for enriching small-molecule active components (e.g., flavonoids, alkaloids) by utilizing the adsorption-desorption effect of resins, but it requires strict control of eluent concentration and flow rate to avoid resin fouling [130]. Modern advanced purification technologies, which are the focus of this study, mainly include chromatographic separation and membrane filtration. Both technologies have been widely applied in the refinement of active ingredients from food and drug homologues in recent years, as they effectively address the limitations of traditional purification methods such as low precision and high solvent residue.

Macroporous adsorption resin technology, as shown in Fig. 1, is an efficient separation and purification method that utilizes the porous structure and huge specific surface area of macroporous adsorption resin to selectively adsorb organic matter in solution through physical adsorption, thereby achieving separation and purification [133]. It is easy to operate, energy-saving, low cost, high product purity, non-hygroscopic [34]. It is widely used in environmental protection, food, medicine and other fields, especially in the extraction, separation and purification of active ingredients in traditional Chinese medicine. Liu et al. [66] used D-3250 resin to separate and purify polysaccharides from safflower bitter beans, and used DEAE Sepharose chromatography column to select different desorption liquids to separate two homogeneous polysaccharides from safflower bitter bean polysaccharides. The application of macroporous adsorption resin technology requires careful selection of suitable resin models and optimisation of the adsorption-desorption conditions to suit the specific characteristics of the substances to be separated, in order to ensure both separation efficiency and product quality.

Chromatographic separation and purification technology is an efficient method for separating and purifying mixtures [43]. This technique is based on the difference between the partition coefficients of different substances in the stationary phase and the mobile phase. By washing the mixture in the mobile phase relative to the stationary phase, different substances move along the stationary phase at different speeds, thereby achieving the purpose of separation and purification. Chromatographic separation and purification technology has the advantages of good selectivity, high separation efficiency, and wide applicability, and is widely used in fields such as pharmaceuticals, food, and chemical engineering [108]. According to the separation mechanism and the difference between stationary phase and mobile phase, chromatographic separation and purification technology can be divided into many types, such as gel chromatography, adsorption chromatography, partition chromatography, ion exchange chromatography, affinity chromatography, etc., to meet the needs of separation and purification of different substances [103]. Gel chromatography (molecular sieve chromatography) uses the molecular sieve effect to separate according to molecular size, which is particularly suitable for the separation of macromolecular substances [70]. Wang et al. [116] used glucan gel G-15 and polyacrylamide gel P2 columns to separate and purify ginseng extract. They obtained a maltooligosaccharide fraction with 2–7 degrees of polymerisation, which can effectively alleviate scopolamine-induced memory impairment in rats. The advantages of chromatographic technology for separation

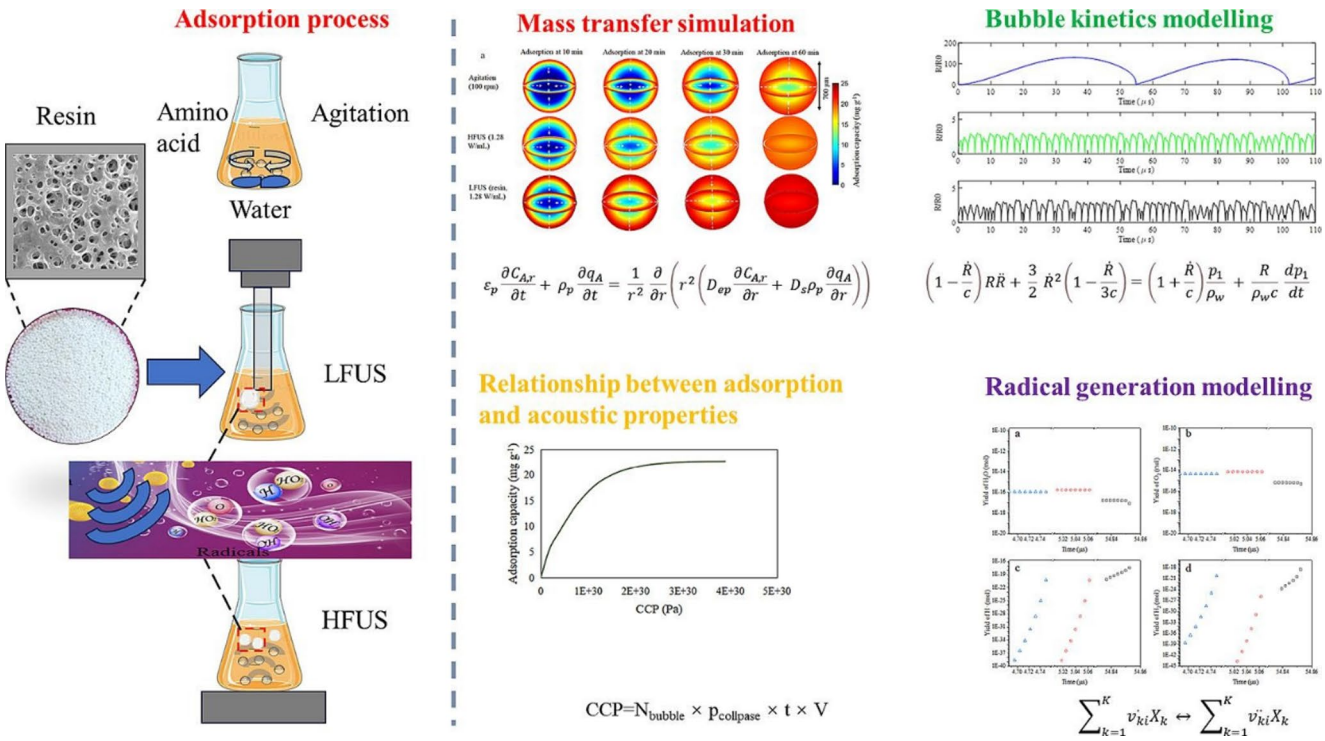

Fig. 1 Technical mechanism of macroporous adsorption resin [124]

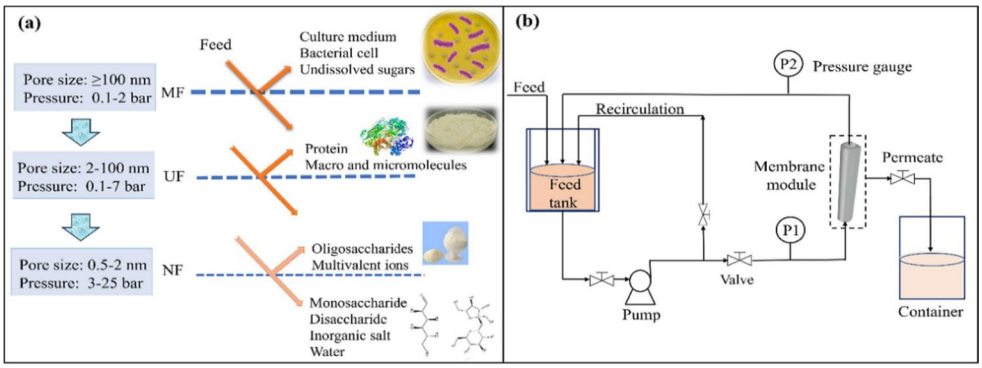

Fig. 2 Membrane separation technologies. **a** Purification process for oligosaccharides and **b** schematic diagram of a separation device [120]

and purification are high separation efficiency and strong selectivity, which can accurately separate target components from complex mixtures, and the separation process can be easily automated and scaled up. Its disadvantages include relatively high equipment and operation costs, in addition, the separation cycle is relatively long, and it has strict requirements on operating conditions (such as temperature and mobile phase ratio) [80].

Membrane separation technology originated in the early 20th century and with the development of polymer material science, high-performance membranes have continuously emerged. Since the 1960 s, this technology has gradually realized industrial application and become an important separation and purification method [100]. Membrane separation technology includes traditional membrane separation technology and new membrane separation technology. Traditional membrane separation technology includes microfiltration, ultrafiltration, nanofiltration, and reverse osmosis, as shown in Fig. 2. Tan et al. [106] successfully developed a novel neutral charge nanofiltration membrane with superhydrophilicity. This nanofiltration membrane has good separation performance for organic molecules with different charges, such as berberine (alkaloids), geniposide (glycosides), sinomenine (alkaloids), and ferulic acid (organic acids). In membrane separation, the separation performance of membranes with similar pore sizes in the same system varies greatly due to the influence of the membrane material itself (surface morphology, hydrophilicity, hydrophobicity, charge, etc.). Li et al. [62] investigated the effects of different ultrafiltration membrane materials on the yield and retention of ginsenosides with a cut-off molecular weight of 100 kDa. The results showed that the cellulose

membrane had a higher retention rate of ginsenosides compared to the polyether sulfone membrane. This is mainly because the hydrophilicity of the two materials is different, and the cellulose membrane has a significantly lower adsorption capacity for ginsenosides. The advantages of membrane separation technology are that it operates at room temperature to protect the activity of components, requires no chemical additives (making it more environmentally friendly), and has high efficiency. Its disadvantages include high cost of membrane materials, limited lifespan and ability to withstand extreme conditions, and the need for pretreatment of high-concentration feed solutions [87].

With the development of technology, new membrane separation technologies include enzyme membrane reactors, intelligent membranes, etc. Enzyme membrane reactors can combine enzymatic catalytic processes with membrane separation processes to achieve simultaneous reaction and separation [105]. Rehman et al. [89] constructed an enzyme membrane reactor using a ceramic ultrafiltration membrane with a molecular weight of 20 kDa, achieving a 63% production of oligofructose with the ability to regulate gut microbiota. The development of new materials provides unlimited possibilities for membrane separation technology. Some new composite materials have the ability to perceive environmental stimuli and respond to perceived information. These materials are known as innovative materials [54]. Smart membranes use the ability of smart materials to perceive and respond to the external environment by changing their own properties, such as membrane pore opening and closing, flux changes, selective differences, etc., to achieve intelligent membrane separation processes [126]. However, at present, the application of smart membranes in the field of food and drug homology is relatively limited.

Fermentation technology

Fermentation technologies for food and drug homologous products can be categorized into traditional fermentation technologies and modern microbial fermentation technologies, both of which aim to transform raw material components, enhance the bioavailability of active substances, and generate new functional metabolites (e.g., probiotics, short-chain fatty acids) to improve product efficacy. Traditional fermentation technologies, which have been used for thousands of years in food processing, mainly rely on natural microbial communities (e.g., mixed strains of lactic acid bacteria, yeast, and mold) for fermentation [18]. Typical applications include the fermentation of goji berries to enhance their antioxidant activity [132]. These technologies have limitations such as unstable fermentation processes, difficulty in controlling product quality, and long fermentation cycles [143]. Modern microbial fermentation technologies, which are the focus of this study, mainly include single-strain directed fermentation and mixed-strain synergistic fermentation. They overcome the shortcomings of traditional methods by relying on clear microbial metabolic pathways and controllable operational parameters, significantly improving the stability of fermentation processes and the consistency of product quality [50]. Moreover, these two technologies can be flexibly adjusted according to the types of food and drug homologous raw materials and the target active ingredients, showing strong applicability in industrial production.

Some plant-based food and drug homologues have their active ingredients mainly present in the cytoplasm and have a relatively high molecular weight, making it difficult to break through the blood-brain barrier. They are usually consumed orally, resulting in a low bioavailability of the active ingredients [67]. After microbial fermentation, enzyme systems such as protease, cellulase, pectinase, amylase, etc. in microorganisms degrade or hydrolyze plant cell walls, exposing or releasing active ingredients in food and drug homologous foods. At the same time, they degrade large molecular substances such as protein, fiber, starch, etc. in plant raw materials into small molecular substances, making them more easily absorbed by the human body, thereby improving therapeutic effects [73]. Single-strain directed fermentation uses screened high-efficiency strains (e.g., Lactobacillus plantarum, Saccharomyces cerevisiae) with clear metabolic pathways to achieve targeted transformation of raw materials [118], for example, the significant enhancement of nutritional components and flavor characteristics of mulberry juice by fermentation with *Lactiplantibacillus plantarum* BXM2 [36]. Mixed-strain synergistic fermentation combines two or more strains with complementary metabolic functions to maximize the utilization of raw material nutrients; for instance, both Bacillus subtilis and Bifidobacterium significantly increased the content of gallic acid in Cornus officinalis fruit culture medium after fermentation [141]. In summary, both technologies address the core pain points of traditional fermentation (such as unstable quality and low efficiency) through precise strain selection and metabolic regulation, providing reliable technical support for the standardized and high-value processing of food and drug homologous raw materials. Their application not only improves the yield and bioavailability of target active ingredients but also lays a foundation for the large-scale industrial production of functional products derived from these raw materials.

Analysis, identification, and detection techniques of functional factors

By identifying functional factors, the efficacy of food and drug homologous products can be scientifically verified, providing consumers with accurate product information and helping enterprises understand the active ingredients in products. This provides a scientific basis for new product development and promotes industrial upgrading. There are many techniques for analyzing, identifying and detecting functional factors in food and drug homologous products, such as spectroscopy, nuclear magnetic resonance, chromatography, and chromatography-mass spectrometry.

Spectroscopy is a method of analyzing the wavelength or intensity of emission, absorption, or scattering generated by the energy level transition of molecules when substances interact with radiant energy. Alfei et al. [3] used Fourier transform infrared spectroscopy to obtain spectra at different concentrations of Gallic acid. The spectral data were selected and a linear regression model was established using the least squares method to analyze Gallic acid in the test sample polymer. The limit of detection of this method was 0.01189 mg/ml, and the limit of quantification was 0.03604 mg/ml. The principle of nuclear magnetic resonance (NMR) is mainly based on the behavior of atomic nuclei in a magnetic field [55]. Through nuclear magnetic resonance technology, the chemical components of food and drug homologous products, including proteins, amino acids, polysaccharides, fatty acids, etc., can be accurately identified. Farag et al. [32] identified seven metabolites in Ceylon cinnamon and Chinese cinnamon using a quantitative metabolomics method based on nuclear magnetic resonance and performed quantitative analysis. The results showed that Ceylon cinnamon had the highest content of eugenol and Chinese cinnamon had the highest content of fatty acids, providing a basis for the study of differences in origin and authenticity of food and drug raw materials. The principle of high-performance liquid chromatography (HPLC) is to use the difference in distribution coefficients of different components in the sample between the stationary phase and the mobile phase to achieve separation [27]. With the advantages of high separation efficiency and sensitivity, HPLC can effectively isolate and identify many active ingredients in food and drug homologous products, such as polyphenols, flavonoids, etc. The principle of mass spectrometry (MS) is that the substance being measured is ionized in an ion source, generating charged ions with different mass to charge ratios. The ions form an ion beam under the action of an accelerating electric field and enter the mass analyzer through a focusing system. The ions are separated based on their mass-to-charge ratio using electric or magnetic fields. As shown in Fig. 3, chromatography-mass spectrometry can be used for qualitative and quantitative analysis, accurately identifying the molecular structure and content of functional factors, ensuring reliable analytical results [37]. Chromatography mass spectrometry technology can provide more comprehensive information on molecular structure and composition. Zhang et al. [137] employed liquid chromatography-mass spectrometry to analyze ginseng fibrous roots and identified the chemical constituents in the total saponin extract from these roots.

The application of modern analytical instruments in functional factor analysis of food and drug homologous products has significantly improved the accuracy and speed of analysis [48]. The application of these technologies has promoted the in-depth development of research on food and drug homologous products, laying a solid foundation for industrial innovation and health protection. It is worth

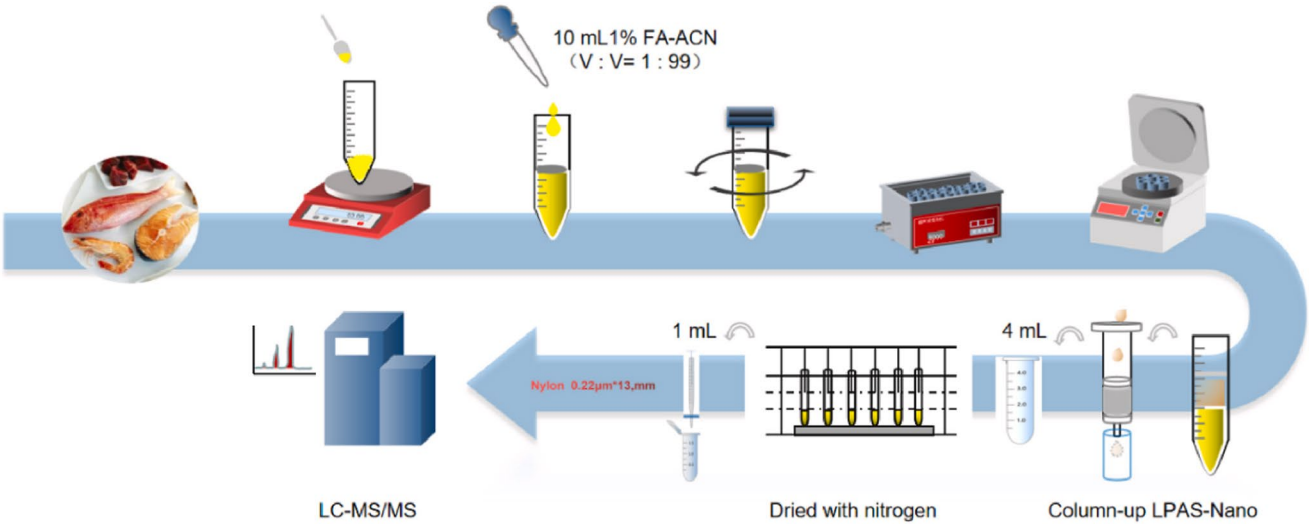

Fig. 3 Simultaneous determination of phenothiazine drugs and their metabolites residues in animal derived foods by high performance liquid chromatography tandem mass spectrometry [39]

mentioning that with the rapid development of science and technology, especially in the 21 st century, various separation, analysis techniques and methods used for the analysis and testing of food and drug homologous products have been continuously improved and updated. Many high-sensitivity and high-resolution analytical instruments have been increasingly applied in the physical and chemical testing of products [92]. At present, while ensuring the precision and accuracy of detection results, the analysis and testing of food and drug homologous products are developing in the direction of micro, rapid, and automated.

New technology for processing food and drug products of the same origin

Other new processing technologies for food and drug homologous products, such as nanotechnology, freeze-drying, and intelligent processing technology, play an important role in food and drug homologous products, as shown in Fig. 4. Nanonization of food and drug homologous components can improve their solubility, stability, and bioavailability, thereby enhancing the efficacy and nutritional value of the products. Freeze drying technology is suitable for dehydration treatment of food and drug homologous products, preserving their nutritional components and active substances, extending their shelf life, and facilitating storage and transportation [52]. Intelligent processing technology can achieve precise control and automated production of food and drug products, ensuring stable product quality, improving production efficiency, and meeting personalized needs. The application of these technologies has promoted the upgrading and development of the food and drug industry, providing consumers with higher quality, safer, and more effective health products.

The principle of nanotechnology is mainly to optimize and enhance material functions by precisely controlling the structure and properties of substances at the nanoscale [53]. The application of nanotechnology can improve the stability and bioavailability of active ingredients. For example, nanoparticulation of drugs can increase their absorption in the gastrointestinal tract [107]. Zhang et al. [135] prepared horsetail algae polysaccharide nanoparticles via the anti-solvent method, which significantly enhanced the nanoparticles' in vitro antioxidant activity. The effective adsorption and sensing of active ingredients in food and drug homologous products by nanomaterials not only promotes the study of pharmacological activity of active compounds, but also plays an important role in their quality control, ensuring the safety of products.

Freeze-drying technology, also known as vacuum freeze-drying technology, is an advanced drying method. The core principle is to freeze the water-containing material in

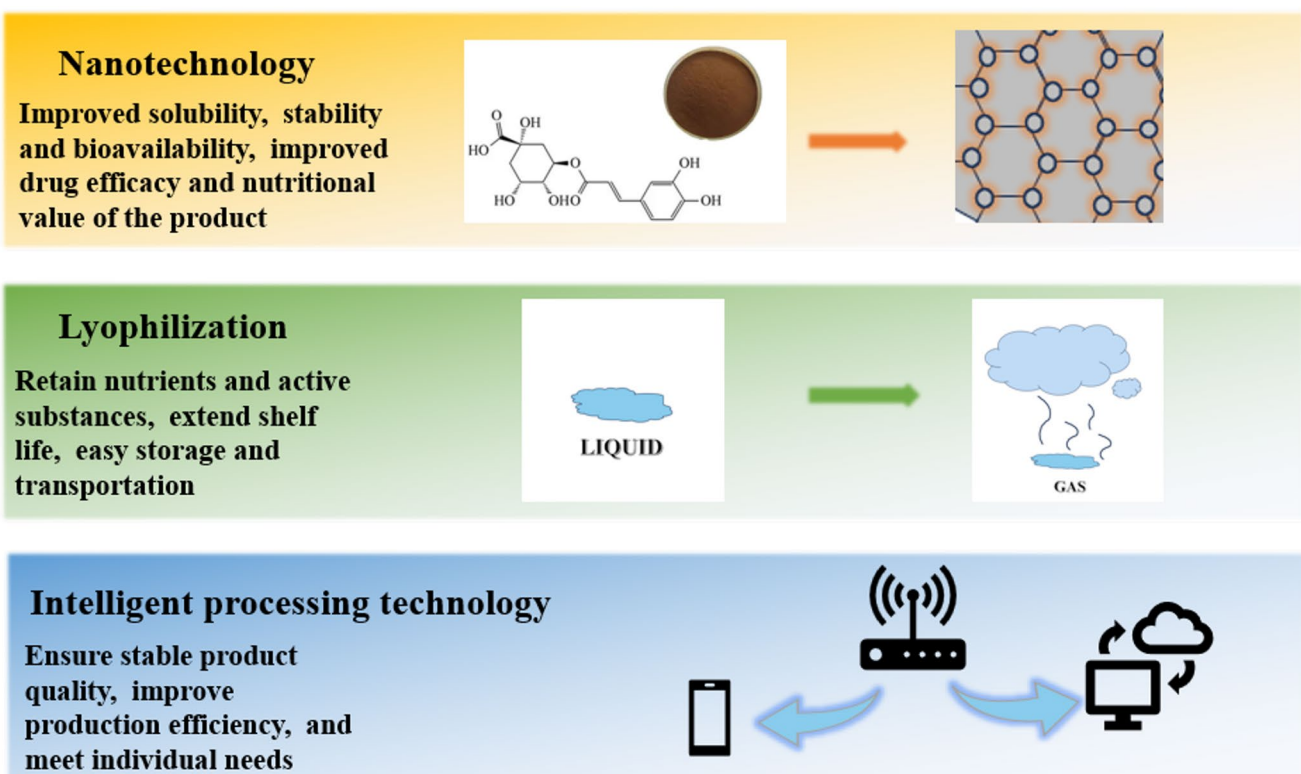

Fig. 4 New processing technology of food and drug homologous products

advance and then sublimate the solid water directly into a gaseous state under vacuum, thereby removing the water from the material [5]. [1] compared the color changes of hawthorn fruit under different drying methods and found that the samples subjected to vacuum freeze-drying had the lowest color change rate due to less contact with oxygen. New types of freeze-drying technology have been developed, such as ultrasound-assisted vacuum freeze-drying technology, microwave-assisted vacuum freeze-drying and pulsated vacuum freeze-drying. Xu et al. [128] studied the effects of different pretreatment methods such as ultrasound on the efficiency and product quality characteristics of vacuum freeze-drying of okra. They found that after ultrasound pretreatment, the total phenolic content and antioxidant capacity of okra were increased, the chlorophyll degradation rate was reduced, and the drying time was shortened and the total energy consumption was reduced. Chen et al. [14] studied the influence of variables such as microwave power density on product quality and energy consumption. They found that the higher the microwave power density, the faster the drying rate. Huang et al. [41] studied the effects of pulse electric field (PEF)-assisted composite freezing mode combined with pre-vacuum freezing on the quality and energy consumption of fresh carrots. They found that pulse vacuum freezing could significantly accelerate the freezing kinetics of conventional freezing (CF) process, and affect the texture, water loss and microstructure of thawed carrots. Freeze-drying technology can preserve the nutritional content, color, flavor, and shape of materials to the maximum extent possible. Especially suitable for thermosensitive substances such as proteins, vitamins, etc., and the activity of microorganisms and enzymes is inhibited during the drying process. However, it is easy to lose some volatile components. An et al. [4] found that the types of volatile components in ginger after vacuum freeze-drying decreased compared to 60 °C hot air drying and infrared drying. Vacuum freeze-drying technology has great potential to improve the quality of dried products and solve the problems of high energy consumption and low efficiency in vacuum freeze-drying. It is widely used in various fields such as medicine, food, and biological products, and has become an indispensable part of modern industry.

The implementation of intelligent processing technology has achieved precise control over the processing process, improving product consistency and quality stability. Intelligent machining technology is an innovative machining mode that integrates advanced technologies such as artificial intelligence, big data, and the Internet of Things to improve machining efficiency, reduce costs, and enhance product quality [76]. Dong et al. [25] used computer vision technology to establish a nonlinear quantitative model by observing the intrinsic relationship between color changes, tea pigments, and sensory scores during the fermentation process of black tea. Among them, the residual predictive deviation (RPD) of the model for theaflavins and sensory scores is greater than 2.5, indicating that the model has good predictive ability for theaflavins and sensory scores, and is expected to implement precise quantitative monitoring of black tea fermentation quality in production. Intelligent processing technology integrates several advanced technologies to achieve automation, precision, and efficiency in the processing process. This technology can significantly improve production efficiency, shorten product introduction time, reduce labor costs, and enhance product quality and competitiveness. As technology advances, intelligent processing technology will continue to evolve, bringing more innovation and change to the food and medical industries.

Quality and safety control and testing technology

The quality and safety control of food and drug homologous products can be strictly controlled through four aspects: raw materials, planting/breeding environment, processing and finished products, as shown in Fig. 5. Raw material suppliers are strictly screened to ensure the traceability of raw material sources and the absence of contamination. Planting/breeding environments are monitored to ensure compliance with ecological standards and to reduce residues of pesticides, heavy metals and other harmful substances, standardized processing procedures are established to ensure that every step meets hygiene and quality requirements, and comprehensive testing of finished products, including nutritional content and residues of harmful substances, is carried out to ensure product safety and efficacy [31].

To ensure the quality and safety of food and drug homologous products, advanced testing techniques should be used to detect pesticide residues, heavy metal contamination, microbial contamination, etc., to ensure that the products comply with relevant standards and regulations [16]. The rapid detection methods for pesticide residues mainly include three categories: biochemical methods, spectroscopic methods, and mass spectrometry methods. Such as enzyme inhibition method, enzyme-linked immunosorbent assay, biosensor method, colloidal gold method, Raman spectroscopy, and near-infrared spectroscopy. Wang et al. [111] detected pesticide residues in honey using liquid chromatography-tandem mass spectrometry and gas chromatography-tandem mass spectrometry techniques. Li et al. [63] used ELISA and semi-quantitative colloidal gold immunochromatography (GICA) to detect the pesticide thiamethoxam, with detection limits of 25.6 and 3.8 ng/mL, respectively. The visual detection limit of the colloidal gold monoclonal antibody probe was 8 ng/mL, which can be detected by naked eye. The detection methods for heavy

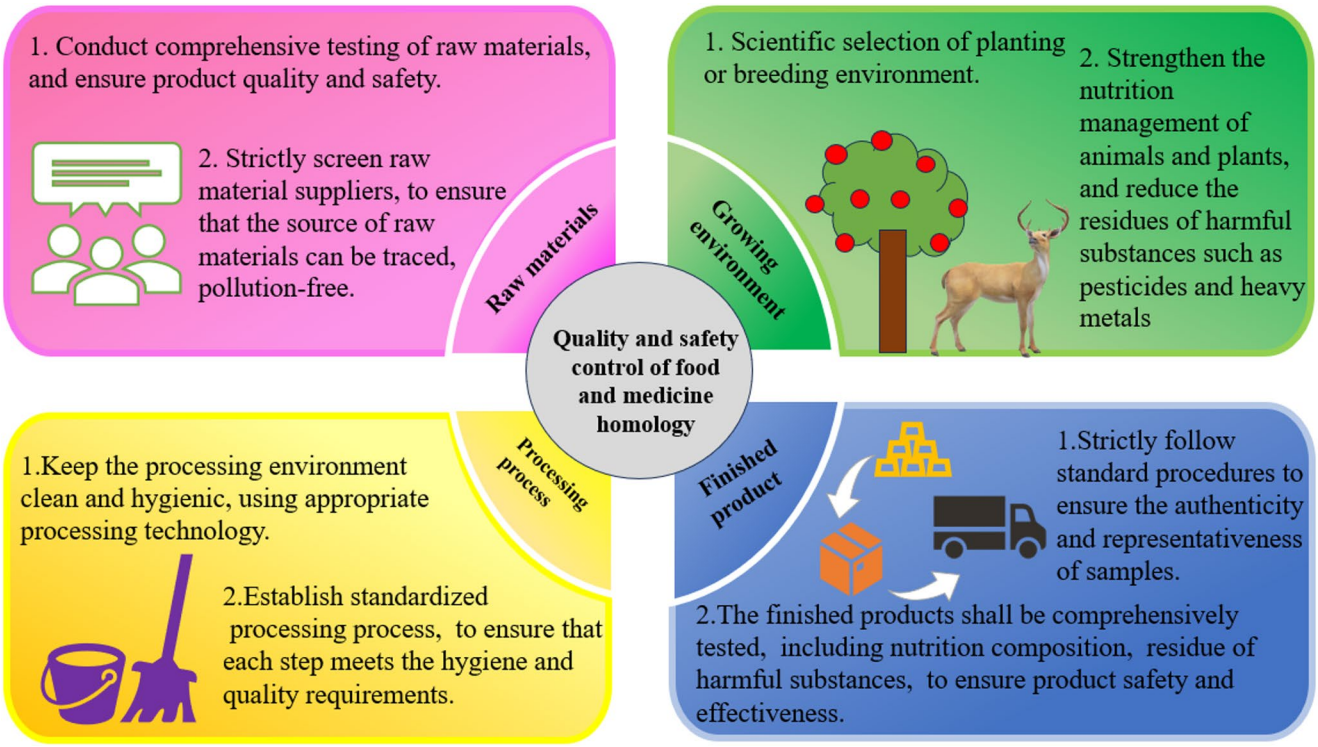

Fig. 5 Food and drug homology quality and safety control

metals include atomic absorption spectroscopy (AAS), inductively coupled plasma mass spectrometry (ICP-MS), X-ray fluorescence spectroscopy (XRF), etc. For example, some scholars have used flame atomic absorption spectroscopy to measure the content of iron, manganese, zinc, and copper in 8 types of Chinese herbal medicines [8]. Some researchers have also used ICP-MS technology to investigate and analyze the heavy metal ion pollution status of *Paeonia lactiflora*, *Angelica sinensis*, *Atractylodes macrocephala* and *Pueraria lobata* [102]. Real time fluorescence quantitative PCR technology can be used to detect microorganisms in food and drug homologous products. By monitoring changes in fluorescence levels in real time, the number of cycles required for the test substance to reach the fluorescence detection threshold can be determined. Okolie [79] used real-time PCR polymorphism analysis to simultaneously detect multiple genes such as marker genes (16 S rRNA), coagulase negative Staphylococcus, Staphylococcus aureus, leukopenin, and methicillin-resistant Staphylococcus, and successfully detected their virulence. Fluorescence quantitative PCR technology can not only accurately identify the types and quantities of pathogenic bacteria in products. Still, it can also detect whether products contain genetically modified ingredients, helping consumers understand the true properties of products and meet the import requirements of specific countries or regions.

Challenges and opportunities of food and drug homologous products

Although the concept of food and drug sharing the exact origin is rooted in the traditional culture of many countries and regions and is widely accepted by people in today's society, there are still many problems and challenges in the selection and acquisition of raw materials, the application and innovation of technology, and the regulation and standardization of the market.

The stability and consistency of raw material supply is a challenge. Due to factors such as climate, soil, and planting/breeding techniques, the yield and quality of raw materials may fluctuate, affecting the quality and stability of products [117]. Enterprises focus on promotion and marketing rather than research and development, and the product form is relatively simple, resulting in a lack of technological competitiveness and large-scale production capacity in the food and drug homology industry. At present, although various levels of government departments have made some efforts in the supervision of food and drug products of the same origin, the overall regulatory efforts are still insufficient [97]. On the one hand, the limited number of regulatory personnel makes it difficult to comprehensively supervise the vast market; on the other hand, the regulatory equipment and technological means are relatively backward, making

it difficult to meet the regulatory needs of food and drug products of the same origin [35].

While there are challenges to the development of food and drug products of the same origin, there are also opportunities. Consumers' increasing attention to health has opened up a wide market for food and drug homologous [45]. Governments around the world have also introduced relevant policies to support the development of the food and drug homologous industry, providing a favorable policy environment for enterprises. Technological innovation has injected new vitality into the development of the food and drug homologous industry. With the continuous development of biotechnology, information technology, etc., more technical means and solutions have been provided for the research and development, production, and quality control of food and drug homologous products. Food and drug-related enterprises should actively respond to the challenges, seize the opportunities, and promote the high-quality development of the industry.

Conclusion and prospect

Food and drug homologous products have a wonderful taste, high nutritional value, safety and no side effects, and have unique advantages in prevention, health care, treatment and other aspects. This article provides a whole chain of raw materials, processing, analysis and quality control. Specific scientific research results are cited to demonstrate the efficacy and functions of food-medicine homologous products, and to clarify the development status of such products. By analyzing the opportunities and challenges of food and drug homologous products, it can be predicted that with the advancement of technology and changes in market demand, food and drug homologous products will meet the personalized needs of consumer groups through innovative product forms, such as functional candies and health tea drinks.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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